Celebal Excellence Internship 2026

Week 4 - Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

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Dataset: Sample - Superstore.csv

Objective

To understand Azure cloud concepts and build a complete end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The assignment covers resource provisioning, pipeline design, metadata validation, IAM role assignment, and pipeline execution monitoring.

Task 1 - Azure Portal & Resource Group

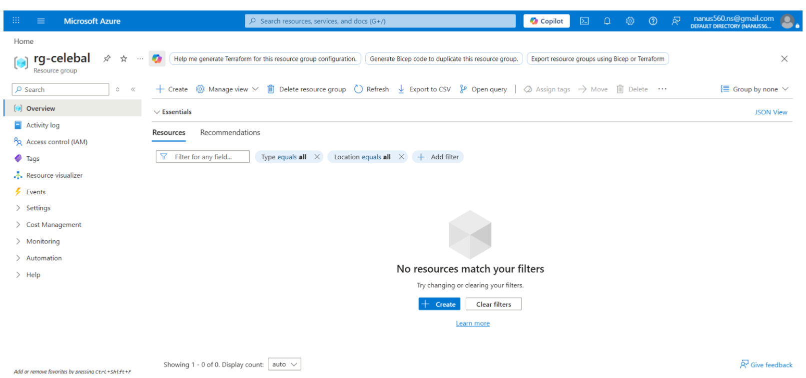
Steps Performed

- Logged into Azure Portal at portal.azure.com
- Navigated to Resource Groups via the search bar
- Clicked + Create and filled in the following configuration:

Subscription	Azure for Students
Resource Group Name	rg-celebal
Region	Central India

- Clicked Review + Create → Create
- Verified Resource Group appeared in the list after deployment

Screenshot-



Task 2 - Storage Account Setup

Steps Performed

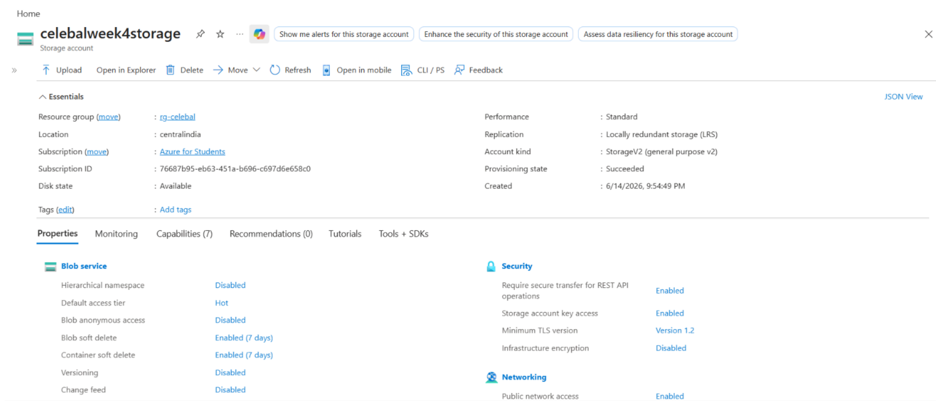
2a. Create Storage Account

- Searched for Storage Accounts in Azure Portal → clicked + Create

Resource Group	rg-celebal-week4
Storage Account Name	celebalweek4storage
Region	Central India

Performance	Standard
Redundancy	LRS (Locally Redundant Storage)

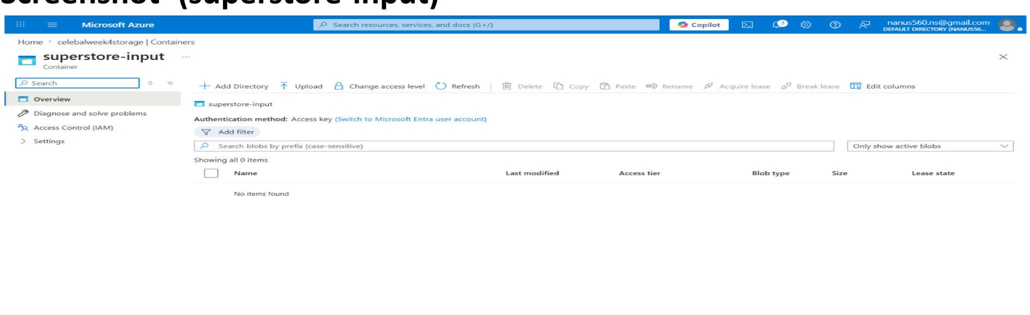
Screenshot-



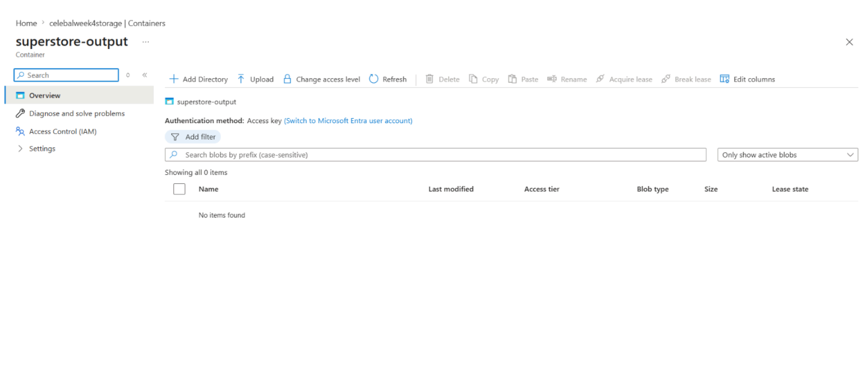
2b. Create Blob Containers

- Inside the Storage Account → navigated to Containers → + Container
- Created container: superstore-input (for source CSV)
- Created container: superstore-output (for processed/destination file)
- Access level kept as Private (no anonymous access)

Screenshot- (superstore-input)



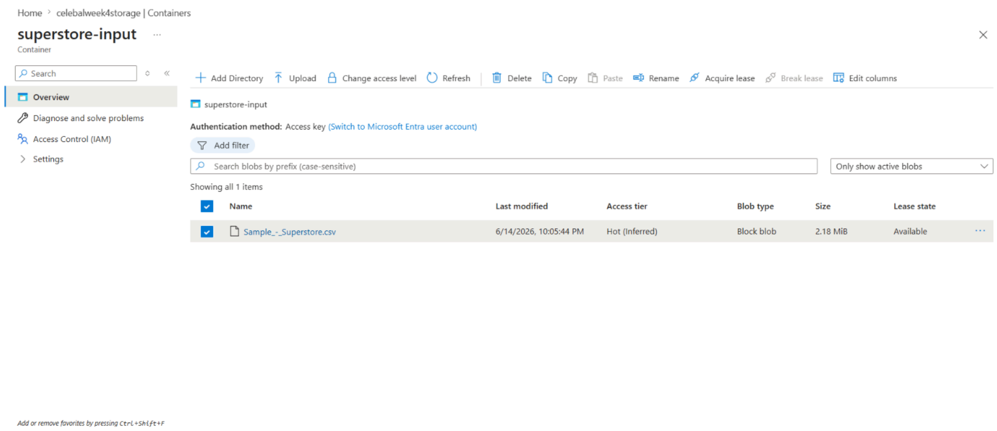
Screenshot- (superstore-output)



2c. Upload CSV File

- Opened superstore-input container → clicked Upload
- Selected and uploaded: Sample_-_Superstore.csv
- Verified file appeared with correct size and last-modified timestamp

Screenshot-



Task 3 - Azure Data Factory Setup

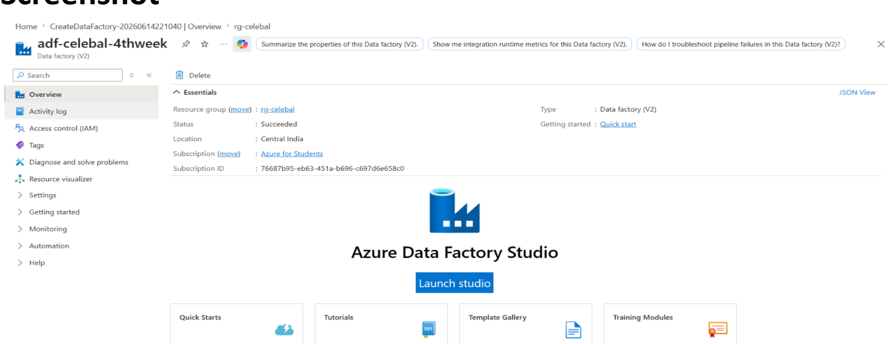
3a. Create Azure Data Factory

- Searched for Data Factories in Azure Portal → clicked + Create

Resource Group	rg-celebal
ADF Name	adf-celebal-4thweek
Region	Central India
Version	V2

- Clicked Review + Create → Create (deployment took ~2-3 minutes)
- After deployment, clicked Launch Studio to open ADF UI

Screenshot-

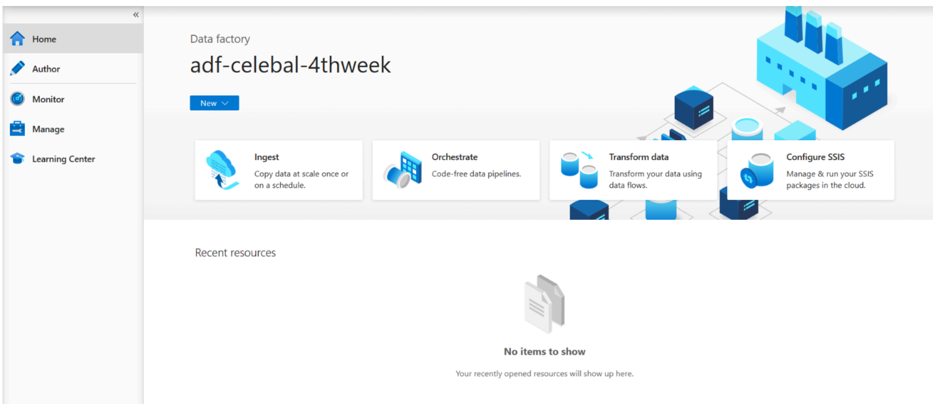


3b. ADF UI Exploration

The ADF Studio consists of three main sections:

Tab	Purpose
Author	Create/edit pipelines, datasets, linked services
Monitor	View pipeline execution status, debug logs, trigger runs
Manage	Manage linked services, integration runtimes, Git config

Screenshot-



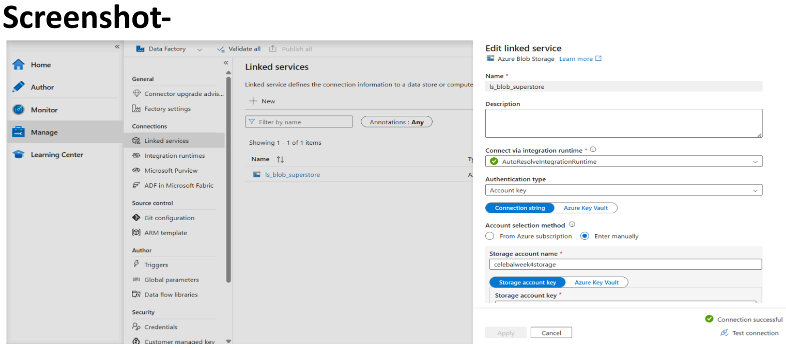
3c. Create Linked Service

- Navigated to Manage tab → Linked Services → + New
- Selected Azure Blob Storage → Continue

Linked Service Name	ls_blob_superstore
Storage Account	celebalweek4storage
Authentication	Account Key (auto-populated)

- Clicked Test Connection → received Connection Successful confirmation
- Clicked Create to save

Screenshot-



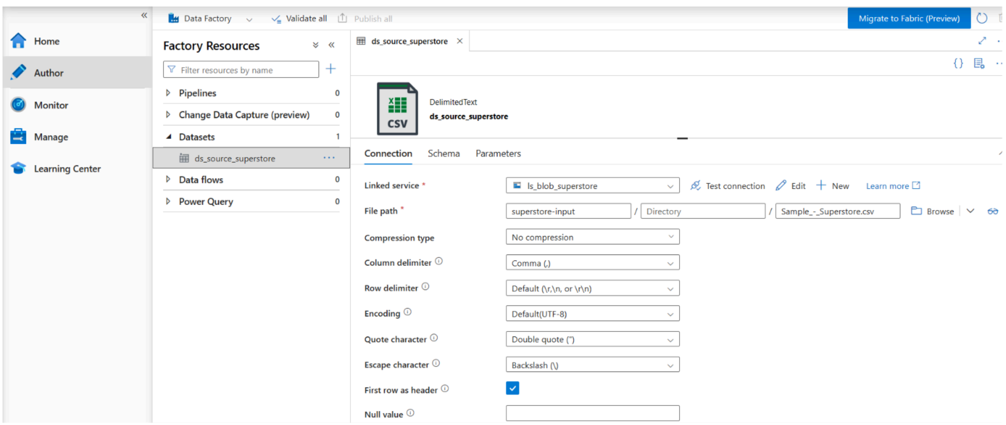
3d. Create Datasets

Source Dataset

- Author tab → Datasets → New Dataset → Azure Blob Storage → DelimitedText

Dataset Name	ds_source_superstore
Linked Service	ls_blob_superstore
Container	superstore-input
File	Sample_-_Superstore.csv
First Row as Header	Yes

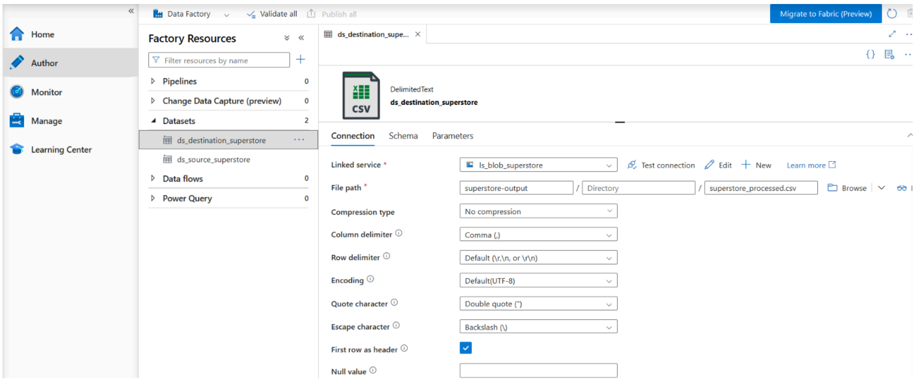
Screenshot-



Destination Dataset

Dataset Name	ds_destination_superstore
Linked Service	ls_blob_superstore
Container	superstore-output
File	superstore_processed.csv
First Row as Header	Yes

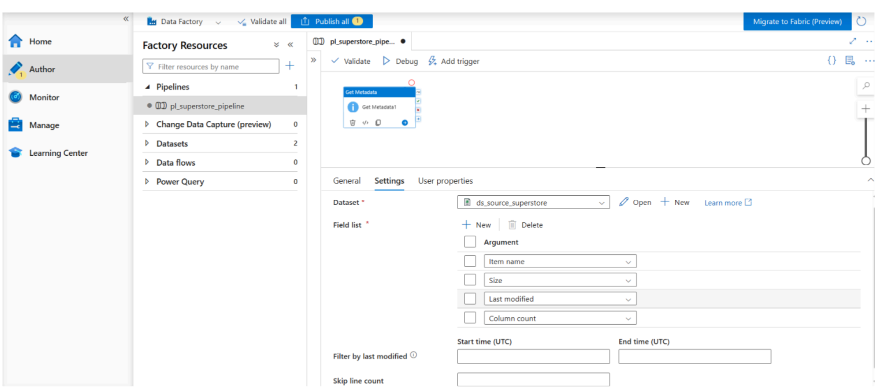
Screenshot-



3e. Get Metadata Activity

- Created a new pipeline: pl_superstore_pipeline
- Dragged Get Metadata activity from the Activities panel onto the canvas
- In Settings tab, selected dataset: ds_source_superstore
- Added the following field list items:
- itemName — retrieves file name
- size — retrieves file size in bytes
- lastModified — retrieves last modification timestamp
- columnCount — retrieves number of columns in the file

Screenshot-



Task 4 - Pipeline Development with Copy Data Activity

Steps Performed

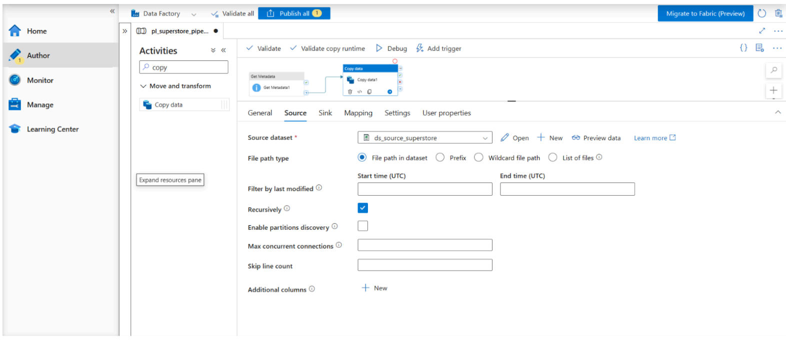
- In the same pipeline pl_superstore_pipeline, dragged Copy Data activity onto the canvas
- Connected Get Metadata → Copy Data using the green success arrow
- This ensures metadata is validated before copying begins

Copy Data Configuration

Source Tab

Dataset	ds_source_superstore

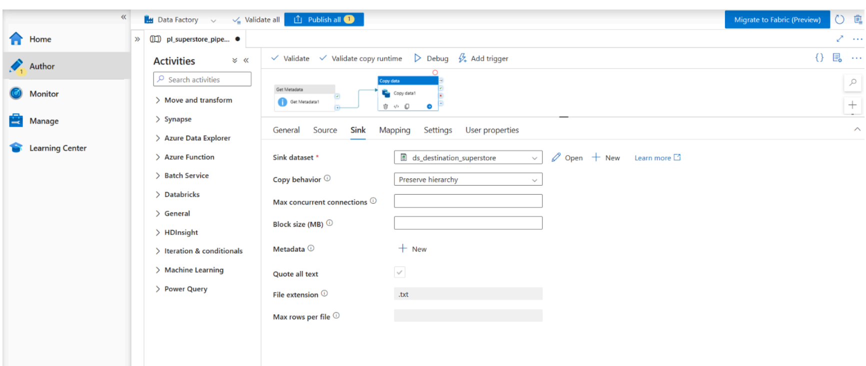
Screenshot-



Sink Tab

Dataset	ds_destination_superstore
Copy Behavior	PreserveHierarchy

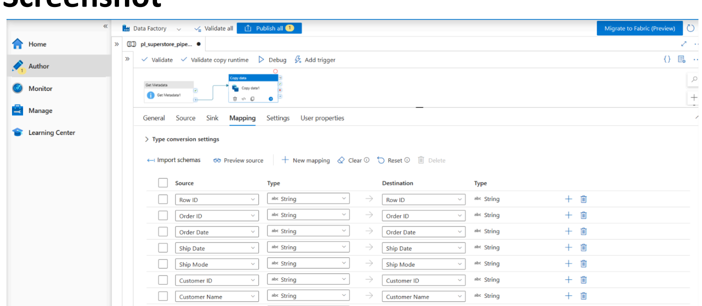
Screenshot-



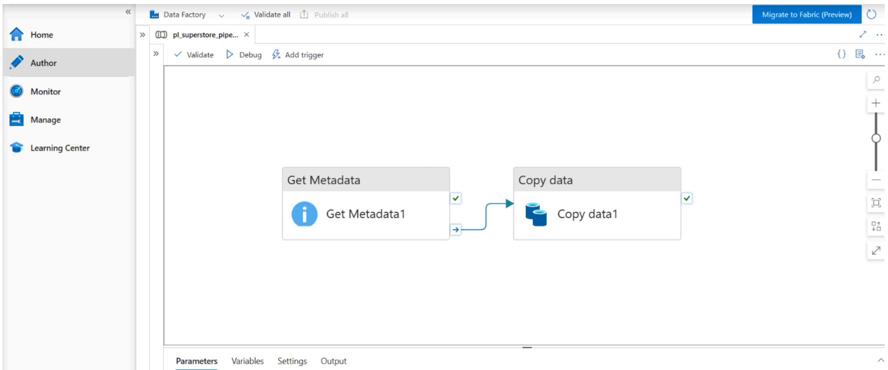
Mapping Tab

- Clicked Import Schemas - all 21 columns from Superstore CSV auto-mapped
- Verified column mapping: Row ID, Order ID, Order Date, Ship Date, ... Profit

Screenshot-



Pipeline Architecture

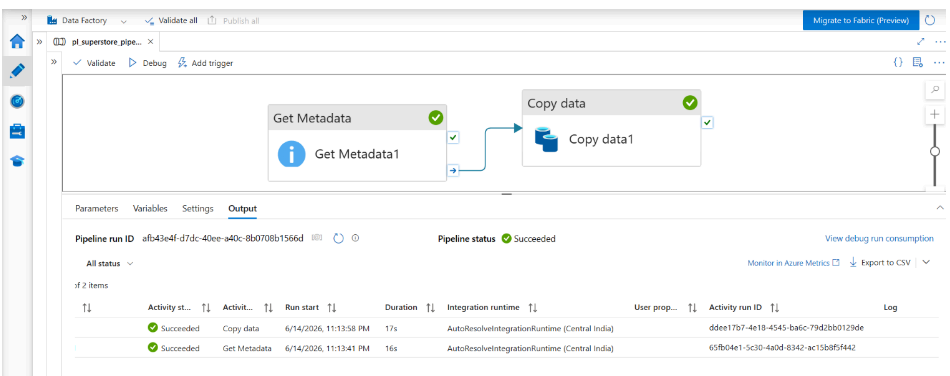


Task 5 - Pipeline Execution

Debug Run

- Clicked Debug button in the pipeline toolbar to run without publishing
- Monitored real-time execution in the Output tab at the bottom.

Screenshot-

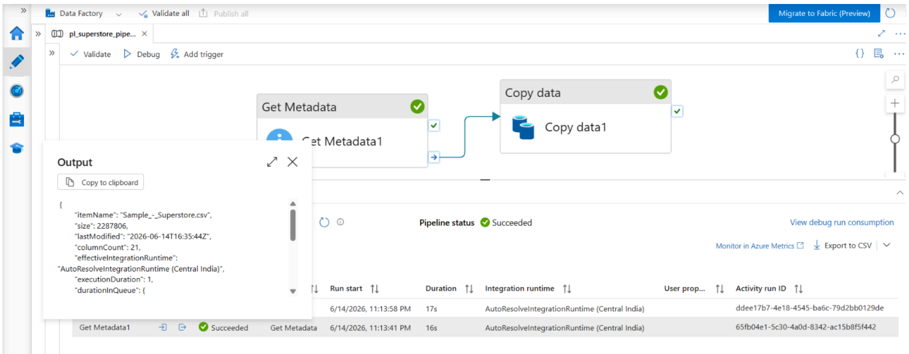


Activity	Status	Result
Get Metadata	Succeeded	Returned itemName, size, lastModified, columnCount
Copy Data	Succeeded	Rows read & written: 9994 | Duration: 17s

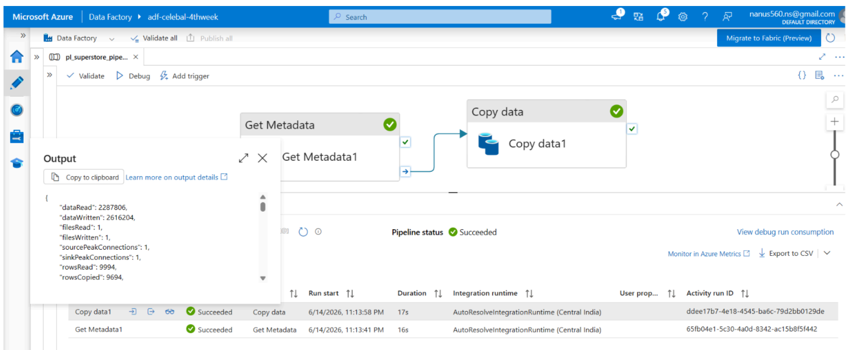
Output Verification

- Navigated to Storage Account → superstore-output container
- Confirmed file superstore_processed.csv was created
- Verified file size matched the source CSV

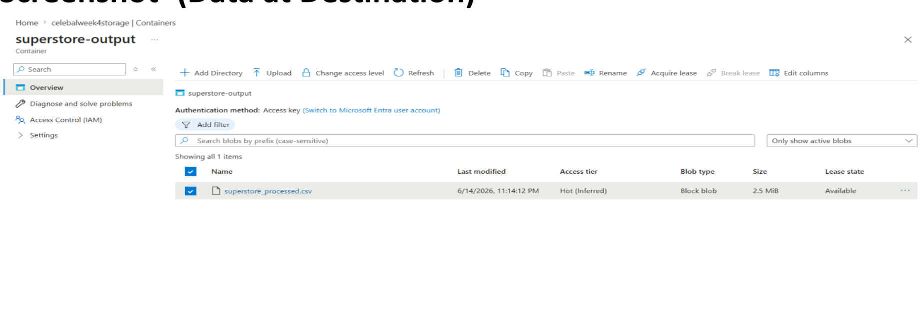
Screenshot-(Get Metadata Output)



Screenshot-(Copy Data Output)



Screenshot- (Data at Destination)



Task 6 - IAM Role Assignment

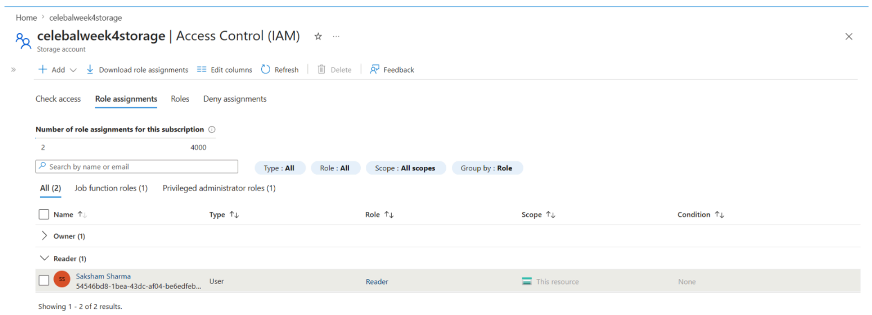
Steps Performed

- Navigated to Storage Account (celebalweek4storage) → Access Control (IAM)
- Clicked + Add → Add role assignment for each of the following:

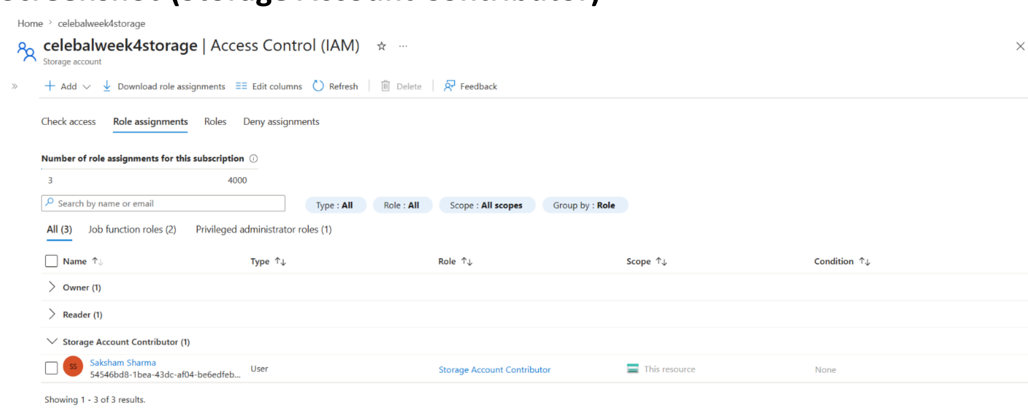
Role Assignments

Role	Assigned To	Purpose
Reader	Saksham (own account)	Read-only access to storage resources
Storage Account Contributor	Saksham (own account)	Create and manage storage resources
Storage Blob Data Contributor	adf-celebal-week4 (Managed Identity)	ADF can read/write to Blob Storage

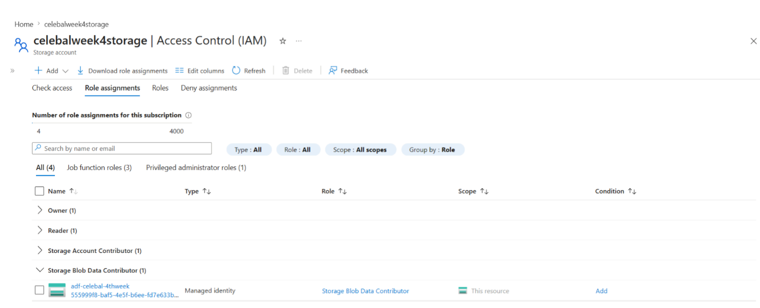
Screenshot-(Reader)



Screenshot-(Storage Account Contributor)



Screenshot-(Get Metadata Output)

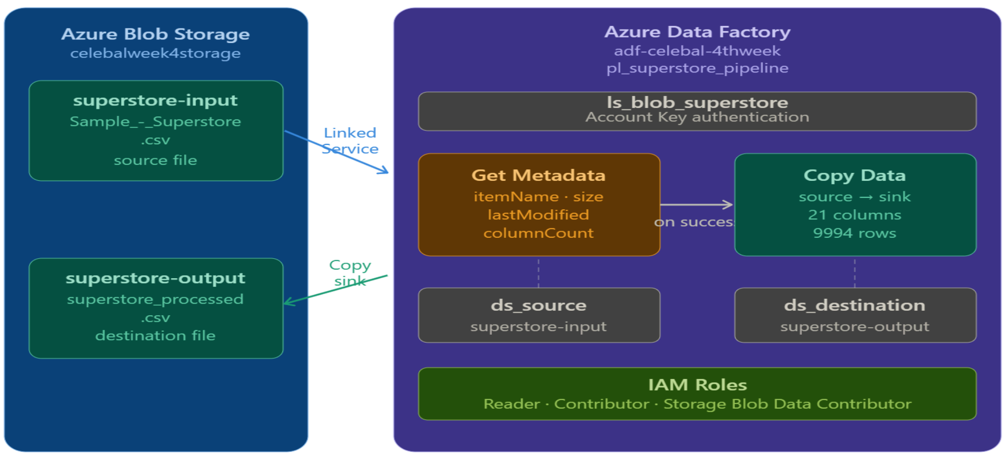


Mini Project - End-to-End Pipeline

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory, copying data from source to destination with metadata validation.

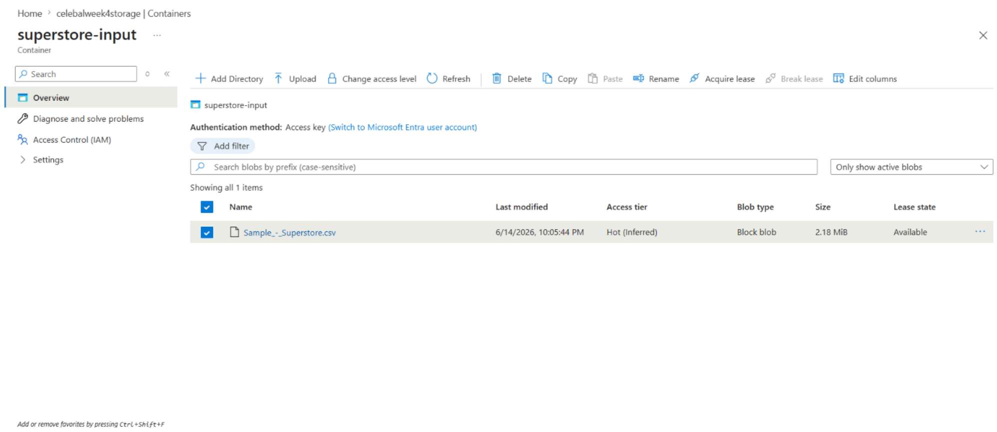
Architecture Overview



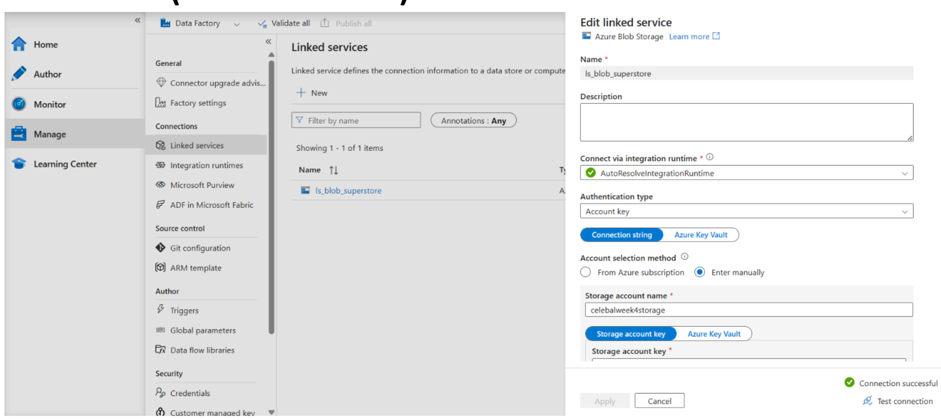
Requirements Fulfilled

Requirement	Implementation
Source: CSV in Blob	Sample_-_Superstore.csv in superstore-input container
Linked Service	ls_blob_superstore (Azure Blob Storage)
Source Dataset	ds_source_superstore (DelimitedText, 21 columns)
Destination Dataset	ds_destination_superstore (superstore-output)
Metadata Check	Get Metadata activity — itemName, size, lastModified, columnCount
Copy Data Activity	Full 21-column copy with schema mapping
Pipeline Executed	Debug run - both activities Succeeded
Data at Destination	superstore_processed.csv created in superstore-output

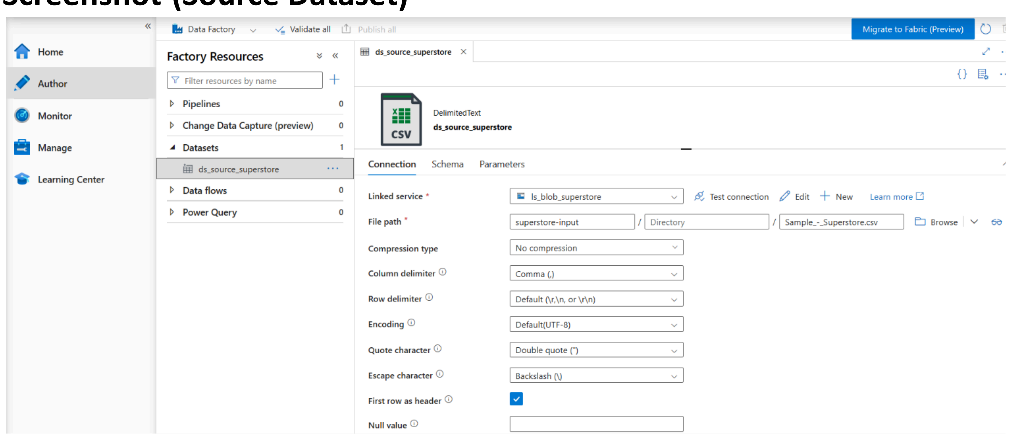
Screenshot-(Source: CSV in Blob)



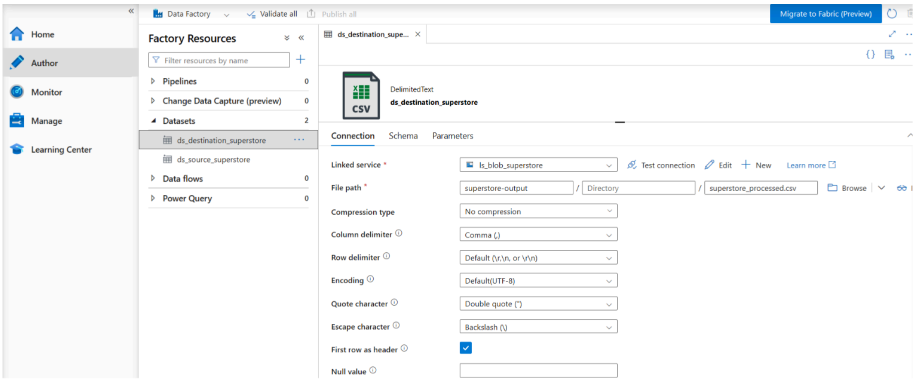
Screenshot-(Linked Services)



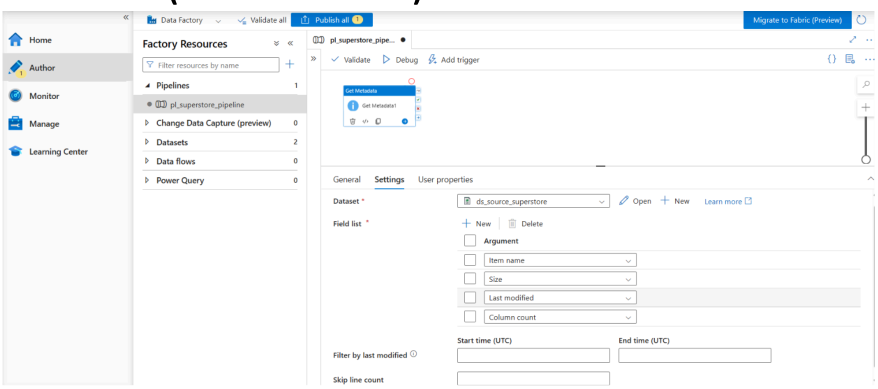
Screenshot-(Source Dataset)



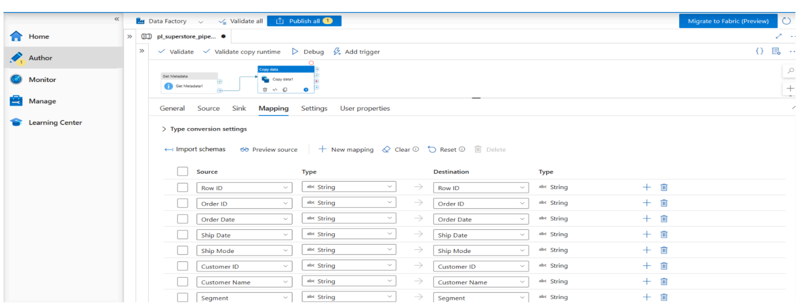
Screenshot-(Destination Dataset)



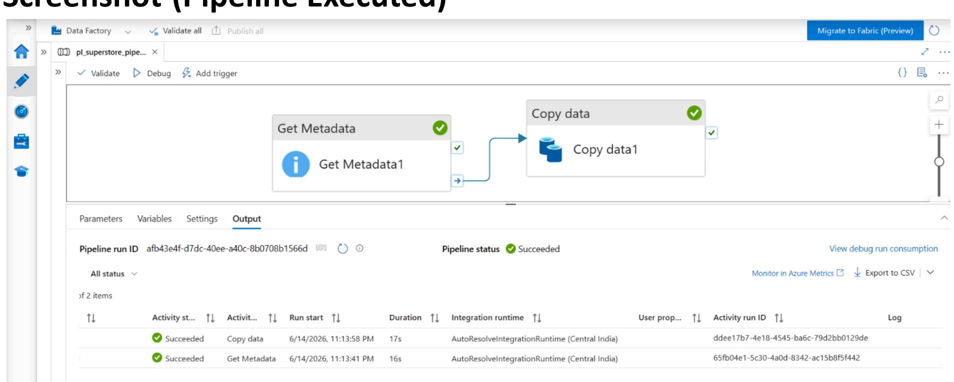
Screenshot-(Metadata Check)



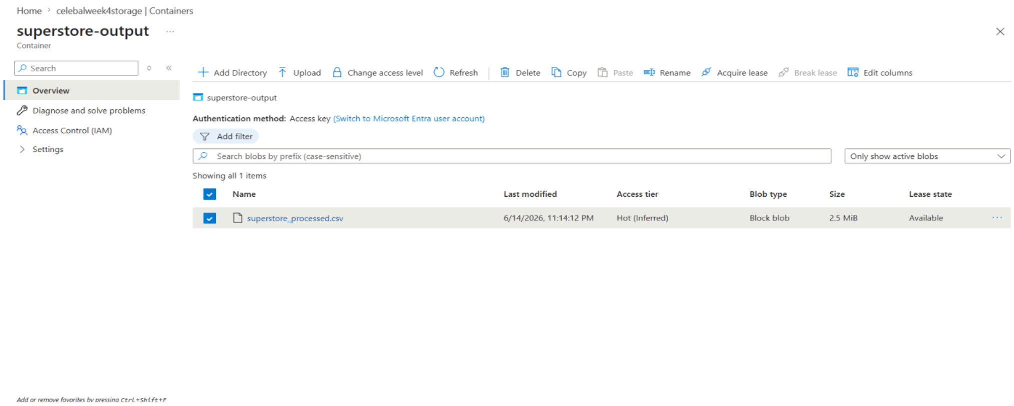
Screenshot-(Copydata Activity)



Screenshot-(Pipeline Executed)



Screenshot-(Data at Destination)



Summary

This assignment provided hands-on experience with Azure cloud services and the Azure Data Factory pipeline ecosystem. Key learnings:

- Set up the Azure environment from scratch by creating a Resource Group, Storage Account, and two Blob Containers (superstore-input and superstore-output) through the Azure Portal.
- Explored the Azure Data Factory Studio and its three core sections: Author (pipeline design), Monitor (run tracking), and Manage (linked services and configuration).
- Configured a Linked Service (ls_blob_superstore) to establish a secure connection between ADF and the Azure Blob Storage account using Account Key authentication.
- Defined two datasets in DelimitedText (CSV) format: ds_source_superstore pointing to the uploaded Superstore file, and ds_destination_superstore for the pipeline output.
- Added a Get Metadata activity to the pipeline that retrieves key file attributes (itemName, size, lastModified, columnCount) from the source CSV before any data movement begins.
- Built the pipeline pl_superstore_pipeline by chaining Get Metadata and Copy Data activities with a success dependency arrow, ensuring metadata is checked before data is moved.
- Configured the Copy Data activity with source and sink datasets, imported the full 21-column schema from the Superstore CSV, and set fault tolerance to handle any malformed rows.
- Ran the pipeline using the Debug trigger and monitored the execution in the Output tab, with both Get Metadata and Copy Data activities completing successfully in under 35 seconds.
- Assigned Reader and Contributor roles to the user account via IAM, and granted the Storage Blob Data Contributor role to the ADF Managed Identity for secure, key-free access to Blob Storage.
- Verified the end-to-end pipeline output by checking the superstore-output container, where superstore_processed.csv was successfully created with all 9,994 rows copied from the source.